

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

Chapter 2: Hilbert Spaces

Based on Bauschke & Combettes

CMS Books in Mathematics, Springer, 2nd ed., 2017

Chapter Overview

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- 2 2.1 Notation and Examples
- 3 2.2 Basic Identities and Inequalities
- 4 2.3 Linear Operators and Functionals
- 5 2.4 Strong and Weak Topologies
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Notation You'll Need: From Chapter 1 to Hilbert Spaces I

Quick Recap from Chapter 1 (Topology and Nets)

Chapter 1 built the topological vocabulary we now reuse without re-deriving it: a **net** $(x_a)_{a \in A}$ is a generalization of a sequence indexed by a directed set, used because in infinite-dimensional spaces sequences alone do not always suffice to characterize closures, continuity, and compactness. A set is **closed** if it contains the limits of all its convergent nets; it is **compact** if every net in it has a subnet converging to a point of the set (equivalently, every open cover has a finite subcover). We also assume familiarity with **Zorn's lemma** and basic normed-space language (norm, Banach space, continuity, Lipschitz continuity).

Notation You'll Need: From Chapter 1 to Hilbert Spaces II

What is a Hilbert Space? (Plain English First)

A **Hilbert space** is a vector space equipped with a notion of *angle and length* — called an **inner product** — that is **complete**, meaning it has no “missing limit points”: every sequence that looks like it should converge (a Cauchy sequence) actually does converge to a point already in the space. Think of it as $\mathbb{R}^n$ generalized to possibly *infinite* dimensions, keeping every piece of Euclidean geometry (Pythagoras, orthogonality, projections) intact.

Notation You'll Need: From Chapter 1 to Hilbert Spaces III

Definition: Inner Product and Hilbert Space

Let $\mathcal{H}$ be a real vector space. An **inner product** (or *scalar product*) is a map $\langle \cdot | \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{R}$ that is bilinear, symmetric ($\langle x | y \rangle = \langle y | x \rangle$), and positive definite ($\langle x | x \rangle > 0$ for $x \neq 0$). It induces the **norm** $\|x\| = \sqrt{\langle x | x \rangle}$ and the **distance** $d(x, y) = \|x - y\|$. $\mathcal{H}$ is a **(real) Hilbert space** if, in addition, $(\mathcal{H}, d)$ is **complete**. Two vectors x, y are **orthogonal**, written $x \perp y$, if $\langle x | y \rangle = 0$ — the abstract version of “perpendicular.”

$\mathbb{R}^n$ as the Motivating Example

In $\mathcal{H} = \mathbb{R}^n$, $\langle x | y \rangle = \sum_{i=1}^n \xi_i \eta_i$ (the familiar dot product), $\|x\| = \sqrt{\sum_i \xi_i^2}$ is Euclidean length, and completeness is just the fact that $\mathbb{R}^n$ has no “holes.” Everything in this chapter is designed so that it reduces to ordinary Euclidean geometry when $\mathcal{H} = \mathbb{R}^n$, while remaining valid when $\mathcal{H}$ is infinite-dimensional (e.g., a space of functions or random variables).

Notation You'll Need: From Chapter 1 to Hilbert Spaces IV

The Running Numeric Example (used throughout this deck)

Throughout, we instantiate every identity with the concrete vectors

$$x = (3, 4) \in \mathbb{R}^2, \quad y = (1, -2) \in \mathbb{R}^2.$$

We have $\|x\| = \sqrt{9 + 16} = 5$, $\|y\| = \sqrt{1 + 4} = \sqrt{5} \approx 2.236$, and $\langle x | y \rangle = 3(1) + 4(-2) = 3 - 8 = -5$.

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Section 2.1: Why a Catalogue of Examples?

Intuition

Before manipulating abstract Hilbert spaces, it helps to know the concrete spaces that occur constantly in applications: finite direct sums, sequence spaces ℓ^2 , matrix spaces, L^2 function spaces, and spaces of random variables. All of them are instances of one recipe: take an inner product, check completeness, done. This section is a reference catalogue you will return to throughout the book.

Hilbert Direct Sum (Example 2.1)

Let I be a nonempty (index) set and $(\mathcal{H}_i, \|\cdot\|_i)_{i \in I}$ a family of real Hilbert spaces. The **Hilbert direct sum** is

$$\bigoplus_{i \in I} \mathcal{H}_i = \left\{ x = (x_i)_{i \in I} \in \prod_{i \in I} \mathcal{H}_i \mid \sum_{i \in I} \|x_i\|_i^2 < +\infty \right\},$$

with coordinatewise addition/scalar multiplication and scalar product $(x, y) \mapsto \sum_{i \in I} \langle x_i \mid y_i \rangle_i$. Here $\prod_{i \in I} \mathcal{H}_i$ is the ordinary Cartesian product; the extra condition $\sum_i \|x_i\|_i^2 < +\infty$ is what keeps the norm finite when I is infinite.

Example Catalogue: $\ell^2(I)$, $\mathbb{R}^N$, Matrix Spaces I

$\ell^2(I)$ and $\mathbb{R}^N$ (Examples 2.2–2.3)

If every $\mathcal{H}_i$ is the Euclidean line $\mathbb{R}$, the direct sum becomes

$$\ell^2(I) = \bigoplus_{i \in I} \mathbb{R}, \quad \langle x \mid y \rangle = \sum_{i \in I} \xi_i \eta_i,$$

the space of square-summable families indexed by I . The **standard unit vectors** $(e_i)_{i \in I}$ are defined by $e_i(j) = 1$ if $j = i$ and 0 otherwise — the same role played by the standard basis vectors of $\mathbb{R}^n$. When $I = \{1, \dots, N\}$, $\ell^2(I)$ is exactly the familiar Euclidean space $\mathbb{R}^N$.

Matrix Spaces (Examples 2.4–2.5)

For strictly positive integers M, N : $\mathbb{R}^{M \times N}$ is the Hilbert space of real $M \times N$ matrices with scalar product $(A, B) \mapsto \text{tra}(A^T B)$ (tra is the trace), inducing the **Frobenius norm** $\|\cdot\|_F$. $\mathbb{S}^N$ is the subspace of $\mathbb{R}^{N \times N}$ consisting of all **symmetric** matrices.

Example Catalogue: $\ell^2(I)$, $\mathbb{R}^N$, Matrix Spaces II

L^p and L^2 Spaces (Examples 2.6–2.8)

Let $(\Omega, \mathcal{F}, \mu)$ be a (positive) measure space. A property holds μ -**almost everywhere** (μ -a.e.) if it holds outside a set of measure zero. For a separable real Hilbert space $(H, \langle \cdot | \cdot \rangle_H)$ and $p \in [1, +\infty[$, $L^p((\Omega, \mathcal{F}, \mu); H)$ is the space of (equivalence classes of) Borel measurable $x : \Omega \rightarrow H$ with $\int_{\Omega} \|x(\omega)\|_H^p \mu(d\omega) < +\infty$; for $p = 2$ it is a real Hilbert space with

$$(x, y) \mapsto \int_{\Omega} \langle x(\omega) | y(\omega) \rangle_H \mu(d\omega).$$

Taking $H = \mathbb{R}$ gives the classical Lebesgue space $L^2(\Omega, \mathcal{F}, \mu)$. Taking $\Omega = [0, T]$ with Lebesgue measure gives $L^2([0, T]; H)$, with scalar product $(x, y) \mapsto \int_0^T \langle x(t) | y(t) \rangle_H dt$.

Random Variables (Example 2.9)

A **probability space** is a measure space $(\Omega, \mathcal{F}, P)$ with $P(\Omega) = 1$; a property holding P -a.e. is said to hold **almost surely** (a.s.). A **random variable** is a measurable $X : \Omega \rightarrow \mathbb{R}$, with expected value $EX = \int_{\Omega} X(\omega) P(d\omega)$ when it exists. Then

$$L^2(\Omega, \mathcal{F}, P) = \{X \text{ r.v.} \mid E|X|^2 < +\infty\}$$

is the Hilbert space of finite-second-moment random variables, with scalar product $(X, Y) \mapsto E(XY)$ — the space in which all of probability-theoretic L^2 (variance, covariance, orthogonality of uncorrelated variables) lives.

Sobolev Space $W^{1,2}$ (Example 2.10)

For $T \in \mathbb{R}_{++}$ and $y \in L^2([0, T]; \mathbf{H})$, the antiderivative $x(t) = x(0) + \int_0^t y(s) ds$ is a.e. differentiable with $x' = y$ a.e. Then $W^{1,2}([0, T]; \mathbf{H}) = \{x \in L^2([0, T]; \mathbf{H}) \mid x' \in L^2([0, T]; \mathbf{H})\}$, with scalar product $(x, y) \mapsto \int_0^T \langle x(t) \mid y(t) \rangle_{\mathbf{H}} dt + \int_0^T \langle x'(t) \mid y'(t) \rangle_{\mathbf{H}} dt$ — functions that are themselves square-integrable *and* whose derivative is too.

Python: Verifying the Running Example's Basic Numbers

We instantiate the Hilbert space $\mathbb{R}^2$ (dot product) and check the norms and inner product of the running example $x = (3, 4)$, $y = (1, -2)$ that we will reuse for every identity in this chapter.

```
1  import numpy as np
2
3  x = np.array([3.0, 4.0])
4  y = np.array([1.0, -2.0])
5
6  inner_xy = np.dot(x, y)           # <x|y>
7  norm_x = np.linalg.norm(x)       # ||x||
8  norm_y = np.linalg.norm(y)       # ||y||
9
10 print(f"<x|y> = {inner_xy}")      # -5.0
11 print(f"||x|| = {norm_x}")        # 5.0
12 print(f"||y|| = {norm_y:.4f}")    # 2.2361
13
14 # ell^2(I) sanity check: standard unit vectors e_1, e_2 are orthonormal
15 e1, e2 = np.array([1.0, 0.0]), np.array([0.0, 1.0])
16 print("e1 . e2 =", np.dot(e1, e2)) # 0.0 (orthogonal)
17 print("||e1|| =", np.linalg.norm(e1)) # 1.0 (unit norm)
```

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Cauchy–Schwarz: Intuition First

Intuition

In $\mathbb{R}^n$, $\langle x | y \rangle = \|x\| \|y\| \cos \theta$ where θ is the angle between x and y . Since $|\cos \theta| \leq 1$ always, we immediately get $|\langle x | y \rangle| \leq \|x\| \|y\|$. The Cauchy–Schwarz inequality is exactly this fact, stated without ever mentioning “angle” explicitly (which is essential once $\mathcal{H}$ is infinite-dimensional and there is no a priori geometric picture). Equality is the case $\cos \theta = \pm 1$, i.e., x and y are parallel.

Fact 2.11 (Cauchy–Schwarz)

Let $x, y \in \mathcal{H}$. Then

$$|\langle x | y \rangle| \leq \|x\| \|y\|.$$

Moreover, $\langle x | y \rangle = \|x\| \|y\| \iff (\exists \alpha \in \mathbb{R}_+) x = \alpha y \text{ or } y = \alpha x$.

Here $\mathbb{R}_+ = [0, +\infty[$ and the condition means x, y point in the same direction (are nonnegative multiples of one another).

Running Example

$|\langle x | y \rangle| = |-5| = 5$ and $\|x\| \|y\| = 5 \times \sqrt{5} \approx 11.18$. Indeed $5 \leq 11.18$, and since $-5 \neq 11.18$ the vectors are *not* positive multiples of each other (their angle is obtuse, consistent with $\langle x | y \rangle < 0$).

Lemma 2.12: The Core Identities — Intuition

Intuition

These four identities are all algebraic consequences of expanding $\langle x \pm y | x \pm y \rangle$, exactly as you would expand $(a \pm b)^2$ in ordinary algebra. Geometrically, (ii) says: in a parallelogram with sides x, y , the sum of the squares of the two diagonals equals twice the sum of the squares of the two sides (a classical Euclidean theorem). (iii) says the inner product is fully recoverable from norms alone — “length determines angle.” (iv) is a three-point refinement used later for convexity arguments.

Lemma 2.12

Let $x, y, z \in \mathcal{H}$. Then:

$$(i) \quad \|x + y\|^2 = \|x\|^2 + 2\langle x | y \rangle + \|y\|^2.$$

$$(ii) \text{ Parallelogram identity: } \|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

$$(iii) \text{ Polarization identity: } 4\langle x | y \rangle = \|x + y\|^2 - \|x - y\|^2.$$

$$(iv) \text{ Apollonius's identity: } \|x - y\|^2 = 2\|z - x\|^2 + 2\|z - y\|^2 - 4\|z - (x + y)/2\|^2.$$

Step-by-Step Proof: Parallelogram and Polarization Identities

Proof of (i)

Just expand using bilinearity and symmetry of $\langle \cdot | \cdot \rangle$:

$$\|x + y\|^2 = \langle x + y | x + y \rangle = \langle x | x \rangle + \langle x | y \rangle + \langle y | x \rangle + \langle y | y \rangle = \|x\|^2 + 2\langle x | y \rangle + \|y\|^2.$$

Deriving (ii) and (iii) from (i)

Replacing y by $-y$ in (i) gives

$$\|x - y\|^2 = \|x\|^2 - 2\langle x | y \rangle + \|y\|^2. \quad (*)$$

Adding (i) and (*): the cross terms $\pm 2\langle x | y \rangle$ cancel their opposite-sign partner only in the *difference*; in the *sum* they double the $\|x\|^2, \|y\|^2$ terms and the $\langle x | y \rangle$ terms cancel:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \text{(ii).}$$

Subtracting (*) from (i): the $\|x\|^2, \|y\|^2$ terms cancel and the $\langle x | y \rangle$ terms add:

$$\|x + y\|^2 - \|x - y\|^2 = 4\langle x | y \rangle \quad \text{(iii).}$$

Numeric Verification of the Parallelogram Law in $\mathbb{R}^2$

Running Example: $x = (3, 4)$, $y = (1, -2)$

$$x + y = (4, 2),$$

$$x - y = (2, 6),$$

$$\text{LHS} = \|x + y\|^2 + \|x - y\|^2 = 20 + 40$$

$$\text{RHS} = 2\|x\|^2 + 2\|y\|^2 = 2(25) + 2(5)$$

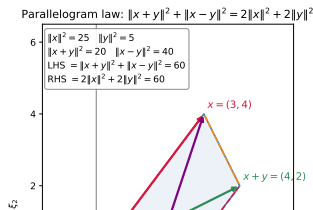
$$\|x + y\|^2 = 16 + 4 = 20,$$

$$\|x - y\|^2 = 4 + 36 = 40,$$

$$= 60,$$

$$= 60. \quad \checkmark$$

Polarization check: $4\langle x | y \rangle = 20 - 40 = -20 \Rightarrow \langle x | y \rangle = -5$, matching the direct computation $3(1) + 4(-2) = -5$. $\checkmark$



Python: Parallelogram, Polarization, and Apollonius — Numeric Check

```
1 import numpy as np
2 rng = np.random.default_rng(1)
3
4 x = np.array([3.0, 4.0])
5 y = np.array([1.0, -2.0])
6 z = np.array([0.0, 0.0])    # arbitrary third point for Apollonius
7
8 lhs_pll = np.dot(x+y, x+y) + np.dot(x-y, x-y)
9 rhs_pll = 2*np.dot(x, x) + 2*np.dot(y, y)
10 print("Parallelogram:", lhs_pll, "==", rhs_pll)           # 60 == 60
11
12 polarization = (np.dot(x+y, x+y) - np.dot(x-y, x-y)) / 4
13 print("Polarization <x|y> recovered:", polarization,      # -5.0
14       " direct <x|y> =", np.dot(x, y))                  # -5.0
15
16 lhs_apo = np.dot(x-y, x-y)
17 rhs_apo = (2*np.dot(z-x, z-x) + 2*np.dot(z-y, z-y)
18           - 4*np.dot(z-(x+y)/2, z-(x+y)/2))
19 print("Apollonius:", lhs_apo, "==", rhs_apo)             # 40 == 40
20
21 # Random stress-test over many (x,y,z) triples in R^5
22 ok = True
23 for _ in range(10000):
24     xr, yr, zr = rng.normal(size=(3, 5))
25     ok &= np.isclose(np.dot(xr+yr, xr+yr)+np.dot(xr-yr, xr-yr),
26                     2*np.dot(xr, xr)+2*np.dot(yr, yr))
27 print("Parallelogram holds for 10000 random R^5 triples:", bool(ok))
```

Lemmas 2.13–2.14: Orthogonality and Barycenters

Intuition for Lemma 2.13

Item (i) says $\langle x | y \rangle \leq 0$ (the angle between x, y is obtuse or right) exactly when moving from x towards $-y$ (i.e., to $x - \alpha y$ for small $\alpha > 0$) never decreases the length $\|x\|$ — a calculus-free way to say “ y is not a descent direction for $\|\cdot\|^2$ at x .” Item (ii) is the same idea specialized to genuine orthogonality, allowing α to range over all of $[-1, 1]$.

Lemma 2.13

Let $x, y \in \mathcal{H}$. Then:

$$(i) \quad \langle x | y \rangle \leq 0 \iff (\forall \alpha \in \mathbb{R}_+) \|x\| \leq \|x - \alpha y\| \iff (\forall \alpha \in [0, 1]) \|x\| \leq \|x - \alpha y\|.$$

$$(ii) \quad x \perp y \iff (\forall \alpha \in \mathbb{R}) \|x\| \leq \|x - \alpha y\| \iff (\forall \alpha \in [-1, 1]) \|x\| \leq \|x - \alpha y\|.$$

Lemma 2.14: Barycentric (Jensen-Type) Identity

For finite families $(x_i)_{i \in I}, (u_i)_{i \in I}$ in $\mathcal{H}$ and weights $(\alpha_i)_{i \in I}$ in $\mathbb{R}$ with $\sum_i \alpha_i = 1$:

$$\left\| \sum \alpha_i x_i \right\|^2 + \frac{1}{2} \sum \sum \alpha_i \alpha_j \|x_i - x_j\|^2 = \sum \alpha_i \|x_i\|^2.$$

Uniform/Strict Convexity and the Hardy–Littlewood–Pólya Inequality

Corollaries 2.15–2.16: Hilbert Spaces are Uniformly and Strictly Convex

For $x, y \in \mathcal{H}$, $\alpha \in \mathbb{R}$:

$$\|\alpha x + (1 - \alpha)y\|^2 + \alpha(1 - \alpha)\|x - y\|^2 = \alpha\|x\|^2 + (1 - \alpha)\|y\|^2.$$

Consequently, if $x \neq y$, $\|x\| = \|y\|$, and $\alpha \in]0, 1[$, then $\|\alpha x + (1 - \alpha)y\| < \|x\|$: *the midpoint (or any strict convex combination) of two distinct equal-norm points is strictly shorter* — the geometric heart of why balls in Hilbert space are “round” with no flat spots, which later drives uniqueness of best approximations (Chapter 3).

Fact 2.18 (Hardy–Littlewood–Pólya)

Let $x, y \in \mathbb{R}^N$ and let $x_{\downarrow}, y_{\downarrow}$ be their *rearrangement vectors* with entries sorted in decreasing order. Then

$$\langle x \mid y \rangle \leq \langle x_{\downarrow} \mid y_{\downarrow} \rangle,$$

with equality iff some permutation matrix P satisfies $Px = x_{\downarrow}$ and $P y = y_{\downarrow}$. *Intuition: pairing up the largest entries of x with the largest entries of y maximizes the dot product — “sorted alignment beats any other pairing.”*

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Bounded Operators: Intuition and Definitions

Intuition

A linear map between normed spaces is continuous exactly when it cannot stretch *any* vector by an unbounded factor; that worst-case stretching factor is the **operator norm**. This mirrors how, for matrices, the largest singular value tells you the maximum possible amplification of any input vector.

Definition: $\mathcal{B}(\mathcal{X}, \mathcal{Y})$ and Operator Norm

For real normed spaces $\mathcal{X}, \mathcal{Y}$,

$$\mathcal{B}(\mathcal{X}, \mathcal{Y}) = \{T : \mathcal{X} \rightarrow \mathcal{Y} \mid T \text{ linear and continuous}\}, \quad \mathcal{B}(\mathcal{X}) = \mathcal{B}(\mathcal{X}, \mathcal{X}),$$

equipped with the norm $\|T\| = \sup_{\|x\| \leq 1} \|Tx\|$. $\mathcal{B}(\mathcal{X}, \mathcal{Y})$ is a normed space, and a Banach space if $\mathcal{Y}$ is.

Example 2.19: Spectral Norm

If $A \in \mathbb{R}^{M \times N}$, then $A \in \mathcal{B}(\mathbb{R}^N, \mathbb{R}^M)$ and its operator norm $\|A\|_2$ (the **spectral norm**) is its largest singular value; moreover $\|A\|_2 \leq \|A\|_F$ (the Frobenius norm bounds the spectral norm).

Continuity, Bilinear Maps, and the Uniform Boundedness Principle

Fact 2.20 and Fact 2.21

A linear $T : \mathcal{X} \rightarrow \mathcal{Y}$ is continuous at *one* point iff it is **Lipschitz continuous** everywhere (linearity turns “continuous somewhere” into “uniformly controlled everywhere”). A bilinear $T : \mathcal{X} \oplus \mathcal{Y} \rightarrow \mathcal{Z}$ is continuous iff $(\exists \beta \in \mathbb{R}_+)(\forall x, y) \|T(x, y)\| \leq \beta \|x\| \|y\|$ — the two-variable analogue of Lipschitz continuity.

Lemma 2.22 (Uniform Boundedness Principle / Banach–Steinhaus)

Let $\mathcal{X}$ be a real Banach space, $\mathcal{Y}$ a real normed space, and $(T_i)_{i \in I}$ operators in $\mathcal{B}(\mathcal{X}, \mathcal{Y})$ that are **pointwise bounded**, i.e., $(\forall x) \sup_{i \in I} \|T_i x\| < +\infty$. Then $(T_i)_{i \in I}$ is **uniformly bounded**: $\sup_{i \in I} \|T_i\| < +\infty$. *Intuition: if a family of continuous linear maps never blows up any single input, completeness of $\mathcal{X}$ forces them to share one common operator-norm bound — no input can be a “hidden” witness of unbounded growth.*

Monotone Operators and the Riesz–Fréchet Representation

Definition 2.23: Monotone Linear Operators

Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be linear and $\alpha \in \mathbb{R}_{++}$. T is:

- **monotone** if $(\forall x \in \mathcal{H}) \langle Tx \mid x \rangle \geq 0$;
- **strictly monotone** if $(\forall x \neq 0) \langle Tx \mid x \rangle > 0$;
- **α -strongly monotone** if $(\forall x) \langle Tx \mid x \rangle \geq \alpha \|x\|^2$.

Intuition: $\langle Tx \mid x \rangle \geq 0$ says T never points “backwards” relative to its input — the operator-theoretic generalization of a nonnegative real number (indeed for $\mathcal{H} = \mathbb{R}$, T monotone just means T is multiplication by a nonnegative scalar). This is the seed concept for the entire theory of monotone operators developed later in the book.

Fact 2.24 (Riesz–Fréchet Representation)

Let $f \in \mathcal{B}(\mathcal{H}, \mathbb{R})$ (a continuous linear functional). Then there exists a *unique* $u \in \mathcal{H}$ such that $(\forall x \in \mathcal{H}) f(x) = \langle x \mid u \rangle$, and $\|f\| = \|u\|$. Every continuous linear functional “is” a vector, via the inner product — this identifies $\mathcal{H}$ with its own continuous dual $\mathcal{H}^*$.

Adjoint Operators, Hyperplanes, and Half-Spaces

Adjoint Operator

If $\mathcal{K}$ is a real Hilbert space and $T \in \mathcal{B}(\mathcal{H}, \mathcal{K})$, the **adjoint** of T is the unique $T^* \in \mathcal{B}(\mathcal{K}, \mathcal{H})$ satisfying $(\forall x \in \mathcal{H})(\forall y \in \mathcal{K}) \langle Tx \mid y \rangle = \langle x \mid T^*y \rangle$. *This generalizes the matrix transpose: for $A \in \mathbb{R}^{M \times N}$, $A^* = A^\top$.*

Fact 2.25 (Selected Adjoint Properties)

With $\ker T = \{x \in \mathcal{H} \mid Tx = 0\}$: (i) $T^{**} = T$; (ii) $\|T^*\| = \|T\| = \sqrt{\|T^*T\|}$; (iv) $(\ker T)^\perp = \overline{\text{ran } T^*}$; (v) $(\text{ran } T)^\perp = \ker T^*$. *These are the abstract analogues of the familiar matrix facts $\text{row}(A)^\perp = \text{null}(A)$ and $\text{col}(A)^\perp = \text{null}(A^\top)$.*

Hyperplanes and Half-Spaces (via Riesz–Fréchet)

For $u \in \mathcal{H} \setminus \{0\}$, $\eta \in \mathbb{R}$: the **closed hyperplane** $\{x \mid \langle x \mid u \rangle = \eta\}$; the **closed half-space** with *outer normal* u is $\{x \mid \langle x \mid u \rangle \leq \eta\}$; the **open half-space** is $\{x \mid \langle x \mid u \rangle < \eta\}$. The distance to the hyperplane $C = \{x \mid \langle x \mid u \rangle = \eta\}$ is

$$d_C(x) = \frac{|\langle x \mid u \rangle - \eta|}{\|u\|}.$$

A Discontinuous Linear Functional (Example 2.27) I

Why This Matters

Fact 2.24 shows every *continuous* linear functional comes from a vector via the inner product. Without continuity this fails badly: using a **Hamel basis** (a purely algebraic, uncountable basis that exists by Zorn's lemma when $\mathcal{H}$ is infinite-dimensional), one can build a linear functional that is discontinuous *everywhere* and whose zero set (a hyperplane) is *dense* in $\mathcal{H}$ rather than closed. This is a cautionary example: linearity alone buys you very little without continuity.

Sketch of the Construction

Gram–Schmidt produces an orthonormal set $B = \{e_n\}_{n \in \mathbb{N}}$ and an uncountable set $C = \{c_a\}_{a \in A}$ with $B \cup C$ a Hamel basis of $\mathcal{H}$. Define $f(x) = \sum_n \xi_n$ where $x = \sum_n \xi_n e_n + \sum_a \eta_a c_a$ (a finite sum). Taking $\alpha_n = 1/(n+1)$ and $x_n = \sum_{k=0}^n \alpha_k e_k$, one shows (x_n) converges strongly to some $z \in \mathcal{H}$ while $f(x_n) \rightarrow +\infty$, so f is discontinuous at z — and by Fact 2.20, discontinuous everywhere. The zero set $C = \{f = 0\}$ is then a hyperplane that is *not* closed (in fact dense, as shown later in Example 8.42).

Python: Operator Norm and Adjoint via a Concrete Matrix

We compute the operator (spectral) norm, the Frobenius norm, and the adjoint (transpose) for a concrete $A \in \mathbb{R}^{2 \times 3}$, verifying $\|A\|_2 \leq \|A\|_F$ (Example 2.19) and $\langle Ax \mid y \rangle = \langle x \mid A^*y \rangle$ (the defining property of the adjoint).

```
1 import numpy as np
2 rng = np.random.default_rng(2)
3
4 A = np.array([[2.0, 0.0, 1.0],
5               [1.0, 3.0, -1.0]])    # A in R^{2x3}, i.e. B(R^3, R^2)
6
7 sing_vals = np.linalg.svd(A, compute_uv=False)
8 spectral_norm = sing_vals.max()    # ||A||_2
9 frobenius_norm = np.linalg.norm(A, 'fro')    # ||A||_F
10
11 print("Singular values:", sing_vals)
12 print(f"||A||_2 = {spectral_norm:.4f} <= ||A||_F = {frobenius_norm:.4f} ?",
13       spectral_norm <= frobenius_norm + 1e-9)
14
15 # Adjoint check: <Ax|y>_{R^2} == <x|A^T y>_{R^3} for random x,y
16 x = rng.normal(size=3)
17 y = rng.normal(size=2)
18 lhs = np.dot(A @ x, y)
19 rhs = np.dot(x, A.T @ y)
20 print(f"<Ax|y> = {lhs:.6f}, <x|A*y> = {rhs:.6f}, equal? {np.isclose(lhs, rhs)}")
```

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Strong Topology vs. Weak Topology: Intuition

Strong (Norm) Topology

The metric topology of $(\mathcal{H}, d)$ is the **strong topology** (or *norm topology*): a net $(x_a)_{a \in A}$ converges strongly to x , written $x_a \rightarrow x$, if $\|x_a - x\| \rightarrow 0$. Used without modifiers, all topological notions in $\mathcal{H}$ (closedness, compactness, etc.) refer to this topology — exactly the topology of “ordinary distance-based convergence” you know from $\mathbb{R}^n$.

Definition 2.29: The Weak Topology

The family of all finite intersections of **open half-spaces** of $\mathcal{H}$ forms the base of the **weak topology**, denoted $\mathcal{H}^{\text{weak}}$. *Intuition: instead of insisting a point is trapped inside a small ball (which needs control of every coordinate/direction at once, hopeless in infinite dimensions), the weak topology only insists a point is trapped on the correct side of finitely many hyperplanes at a time — a much coarser, easier-to-satisfy notion of “closeness.”*

Weak Convergence of Nets

A net $(x_a)_{a \in A}$ **converges weakly** to x , written $x_a \rightharpoonup x$, if $(\forall u \in \mathcal{H}) \langle x_a | u \rangle \rightarrow \langle x | u \rangle$. Every “linear measurement” of x_a converges to the same measurement of x , even though x_a itself may never get close to x in norm.

Lemma 2.30: The Weak Topology is Hausdorff — Step-by-Step Proof

Why This Matters

Hausdorff means distinct points can always be topologically separated; without it, weak limits would not even be unique. The proof is a clean illustration of how half-space neighborhoods are built from the inner product.

Proof

Suppose $x \neq y$ in $\mathcal{H}$. Set $u = x - y$ (a nonzero vector, since $x \neq y$) and $w = (x + y)/2$ (their midpoint). Define

$$U = \{z \in \mathcal{H} \mid \langle z - w \mid u \rangle > 0\}, \quad V = \{z \in \mathcal{H} \mid \langle z - w \mid u \rangle < 0\}.$$

Step 1: U, V are open half-spaces (hence weakly open), since $z \mapsto \langle z - w \mid u \rangle$ is a continuous affine functional. **Step 2:** $x \in U$, because $\langle x - w \mid u \rangle = \langle (x - y)/2 \mid x - y \rangle = \frac{1}{2}\|x - y\|^2 > 0$. Likewise $y \in V$, since $\langle y - w \mid u \rangle = -\frac{1}{2}\|x - y\|^2 < 0$. **Step 3:** $U \cap V = \emptyset$ trivially, since a number cannot be simultaneously > 0 and < 0 . Hence U, V are disjoint weak neighborhoods of x, y respectively, proving $\mathcal{H}^{\text{weak}}$ is Hausdorff. ■

Example 2.32: An Orthonormal Sequence Converges Weakly but Not Strongly

Setup and Intuition

Suppose $\mathcal{H}$ is infinite-dimensional and $(x_n)_{n \in \mathbb{N}}$ is an **orthonormal sequence**: $\|x_n\| = 1$ for all n and $\langle x_n | x_m \rangle = 0$ for $n \neq m$. Intuitively, each x_n “points in a brand-new direction,” so from the point of view of any *fixed* linear measurement u , only finitely much of u ’s “energy” can align with each x_n before it is used up — forcing $\langle x_n | u \rangle \rightarrow 0$. But the vectors themselves never approach each other or 0 in length.

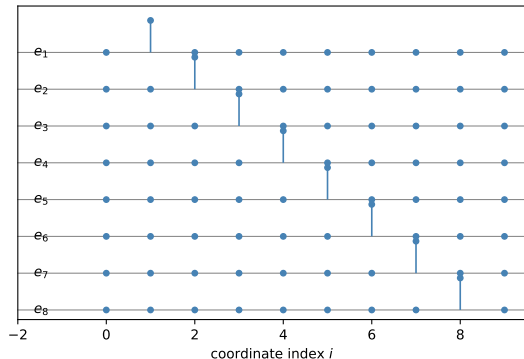
Step-by-Step Proof

Step 1 (Bessel’s inequality): For any $u \in \mathcal{H}$, $\sum_{n \in \mathbb{N}} |\langle x_n | u \rangle|^2 \leq \|u\|^2 < +\infty$. **Step 2:** A convergent series has terms $\rightarrow 0$, so $\langle x_n | u \rangle \rightarrow 0$ for every fixed u . By definition, this says $x_n \rightharpoonup 0$. **Step 3:** However $\|x_n\| \equiv 1 \neq 0$, so $x_n \not\rightarrow 0$ strongly. In fact (x_n) has *no* Cauchy subsequence at all: for $n \neq m$, $\|x_n - x_m\|^2 = \|x_n\|^2 + \|x_m\|^2 = 2$ (Pythagoras, using orthogonality), a fixed positive distance that never shrinks. **Conclusion:** the unit sphere $\{x \mid \|x\| = 1\}$ is (strongly) closed but *not* weakly sequentially closed, since it contains every x_n but not their weak limit 0. ■

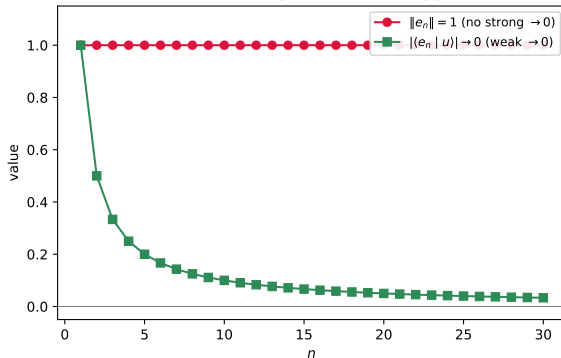
Weak vs. Strong Convergence: Figure

Weak vs. strong convergence of the orthonormal sequence $(e_n)_{n \in \mathbb{N}}$ (Example 2.32)

Orthonormal sequence (e_n) : unit spike
marches off to new coordinates



$e_n \rightarrow 0$ weakly but $e'_n \rightarrow 0$ strongly



Reading the Figure

Left: each standard basis vector e_n places its one unit of “mass” on a coordinate no previous vector

Python: Numerically Confirming Example 2.32

```
1 import numpy as np
2
3 N_dim = 500                                # truncated ambient dimension (stand-in for infinity)
4 u = 1.0 / np.arange(1, N_dim + 1)         # fixed u = (1, 1/2, 1/3, ...) in  $\ell^2(N)$ 
5 print("||u|| =", np.linalg.norm(u))        # finite: u is in  $\ell^2$ 
6
7 def e(n, dim=N_dim):
8     v = np.zeros(dim)
9     v[n - 1] = 1.0
10    return v
11
12 # Weak convergence indicator:  $\langle e_n | u \rangle = u_n \rightarrow 0$ 
13 for n in [1, 5, 20, 100, 300]:
14     inner = np.dot(e(n), u)
15     print(f"n={n:4d}:   $\langle e_n | u \rangle = \{inner:.6f\}$     ||e_n|| = {np.linalg.norm(e(n)):.1f}")
16
17 # No Cauchy subsequence: pairwise distances are all  $\sqrt{2}$ 
18 dists = [np.linalg.norm(e(n) - e(m)) for n, m in [(1, 2), (3, 50), (10, 400)]]
19 print("pairwise ||e_n - e_m|| for distinct n,m:", dists)    # all  $\sqrt{2} = 1.4142...$ 
```

Compactness in the Weak Topology

Fact 2.33: Four Equivalent Characterizations of Finite Dimension

The following are equivalent: (i) $\mathcal{H}$ is finite-dimensional; (ii) the closed unit ball $B(0; 1)$ is (strongly) compact; (iii) the weak topology coincides with the strong topology; (iv) the weak topology is metrizable. *In infinite dimensions, none of these hold — the weak topology is a genuinely different, coarser structure.*

Fact 2.34 (Banach–Alaoglu–Bourbaki)

The closed unit ball $B(0; 1)$ of $\mathcal{H}$ is **weakly compact** — *always*, regardless of dimension. *This is the payoff of weakening the topology: we trade away metrizability/finite-dimensional intuition (Fact 2.33) but recover compactness of bounded sets, which is indispensable for existence proofs (e.g., minimizers of convex functions) throughout the rest of the book.*

Fact 2.35, Lemma 2.36, Fact 2.37 (Eberlein–Šmulian)

The weak topology on $B(0; 1)$ is metrizable iff $\mathcal{H}$ is separable (Fact 2.35). A set $C \subset \mathcal{H}$ is weakly compact iff it is weakly closed and bounded (Lemma 2.36). Weak compactness and weak *sequential* compactness coincide (Fact 2.37) — so despite the weak topology often not being metrizable, sequences still suffice to test weak compactness.

Consequences: Corollary 2.38, Lemma 2.39, Lemma 2.41

Corollary 2.38

For $C \subset \mathcal{H}$, TFAE: (i) C weakly compact; (ii) C weakly sequentially compact; (iii) C weakly closed and bounded. (Combine Lemma 2.36 and Fact 2.37.)

Lemma 2.39

A **bounded** set C is weakly closed iff it is weakly sequentially closed. *Boundedness is essential here* — Remark 2.40 notes that without it, weakly sequentially closed sets need not be weakly closed (Example 3.33 later in the book).

Lemma 2.41

If $\mathcal{K}$ is a real Hilbert space and $T : \mathcal{H} \rightarrow \mathcal{K}$ is a **continuous affine** operator, then T is **weakly continuous**. *Proof idea:* write $T = L + T0$ with L linear and continuous; for a net $x_a \rightharpoonup x$ and any $y \in \mathcal{K}$, $\langle Lx_a \mid y \rangle = \langle x_a \mid L^*y \rangle \rightarrow \langle x \mid L^*y \rangle = \langle Lx \mid y \rangle$, i.e., $Lx_a \rightharpoonup Lx$, so $Tx_a = T0 + Lx_a \rightharpoonup T0 + Lx = Tx$.

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Lemma 2.42: The Norm is Weakly Lower Semicontinuous — Proof

Intuition

Weak convergence can “lose length”: a weak limit can be *shorter* than what the sequence’s lengths suggest (as in Example 2.32, where $\|x_n\| \equiv 1$ but the weak limit 0 has norm $0 < 1$). What can never happen is the reverse — the limit *surprising you by being longer*. This one-sided control is **weak lower semicontinuity** of the norm.

Lemma 2.42

For every net $(x_a)_{a \in A}$ in $\mathcal{H}$ and every $x \in \mathcal{H}$:

$$x_a \rightharpoonup x \quad \Rightarrow \quad \|x\| \leq \underline{\lim} \|x_a\|.$$

Proof

Suppose $x_a \rightharpoonup x$. By definition of weak convergence (taking $u = x$), $\langle x_a | x \rangle \rightarrow \langle x | x \rangle = \|x\|^2$. By Cauchy–Schwarz, $|\langle x_a | x \rangle| \leq \|x_a\| \|x\|$. Hence

$$\|x\|^2 = \lim \langle x_a | x \rangle = \underline{\lim} \langle x_a | x \rangle \leq \underline{\lim} (\|x_a\| \|x\|) = (\underline{\lim} \|x_a\|) \|x\|.$$

If $x = 0$, the conclusion is trivial ($0 \leq \lim \|x_a\|$). Otherwise, divide both sides by $\|x\| > 0$ to get

Lemmas 2.43–2.44: Joint Continuity Along Mixed Nets

Intuition

A bilinear operator that is norm-continuous stays continuous when one argument converges weakly and the other strongly, *provided* the weakly-converging net stays bounded. Boundedness compensates for the weaker mode of convergence — this is the recurring pattern throughout weak-convergence analysis: weak convergence alone is fragile, but “weak + bounded” behaves almost as well as strong convergence.

Lemma 2.43

Let $\mathcal{G}, \mathcal{K}$ be real Hilbert spaces and $T : \mathcal{H} \oplus \mathcal{G} \rightarrow \mathcal{K}$ bilinear with $\|T(x, u)\| \leq \beta \|x\| \|u\|$ for some $\beta \in \mathbb{R}_+$. If (x_a) is **bounded** with $x_a \rightharpoonup x$, and $u_a \rightarrow u$ (strongly), then $T(x_a, u_a) \rightarrow T(x, u)$.

Lemma 2.44 (Special Case: the Inner Product Itself)

If (x_a) is bounded with $x_a \rightharpoonup x$ and $u_a \rightarrow u$ strongly, then $\langle x_a | u_a \rangle \rightarrow \langle x | u \rangle$ (as an ordinary limit in $\mathbb{R}$, i.e., apply Lemma 2.43 with $\mathcal{K} = \mathbb{R}$, $T = \langle \cdot | \cdot \rangle$).

Lemmas 2.45–2.46: Weakly Convergent Subsequences and Uniqueness

Lemma 2.45

Every **bounded** sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathcal{H}$ possesses a **weakly convergent subsequence**. *Proof idea:* let $\rho = \sup_n \|x_n\|$; $B(0; \rho)$ is weakly sequentially compact by Banach–Alaoglu–Bourbaki (Fact 2.34) together with Eberlein–Šmulian (Fact 2.37), and (x_n) lies inside it.

Lemma 2.46

A sequence $(x_n)_{n \in \mathbb{N}}$ **converges weakly** iff it is **bounded** and has **at most one** weak sequential cluster point. *Intuition:* this is the weak-topology analogue of the familiar fact that a bounded real sequence converges iff it has a unique cluster point — boundedness guarantees some cluster point exists (Lemma 2.45); uniqueness then pins down which one is the limit.

Lemma 2.47

If, for every x in a set C , $(\|x_n - x\|)_n$ converges and every weak sequential cluster point of (x_n) lies in C , then (x_n) converges weakly to a point of C . *A convenient sufficient condition for weak convergence used heavily in fixed-point/Fejér-monotonicity arguments later in the book (Chapter 5).*

Propositions 2.48–2.50: Relating Sequences to Orthonormal Systems

Proposition 2.48

If (y_n) is orthonormal and $x_n - y_n \rightarrow 0$ (strongly), then $x_n \rightharpoonup 0$. *Follows immediately from Example 2.32, since $y_n \rightharpoonup 0$ and subtracting a strongly-vanishing perturbation does not change the weak limit.*

Proposition 2.49 (Partial Converse)

If $x_n \rightharpoonup 0$ and $\|x_n\| \equiv 1$, then some subsequence (x_{k_n}) and some orthonormal sequence (y_n) satisfy $x_{k_n} - y_n \rightarrow 0$. *So every unit-norm, weakly-null sequence “looks like” an orthonormal sequence up to a strongly-vanishing correction, along a subsequence — a converse to Example 2.32 / Proposition 2.48.*

Proposition 2.50

If $\overline{\text{span}}\{e_i\}_{i \in I} = \mathcal{H}$ (a **total** or *spanning* family), then $x_n \rightharpoonup x$ iff (x_n) is bounded and $\langle x_n | e_i \rangle \rightarrow \langle x | e_i \rangle$ for every single $i \in I$. *Weak convergence is testable on any spanning/total family of test vectors, not just “all of $\mathcal{H}$ ” — a major practical simplification, e.g. testing against a countable orthonormal basis suffices in a separable Hilbert space.*

Lemma 2.51 and Corollary 2.52: When Weak Becomes Strong

Lemma 2.51

- (i) $x_n \rightharpoonup x$ and $\overline{\lim} \|x_n\| \leq \|x\| \iff x_n \rightarrow x$ (strongly).
- (ii) If $\mathcal{H}$ is **finite-dimensional**, $x_n \rightharpoonup x \iff x_n \rightarrow x$ (weak and strong topologies coincide, as already seen in Fact 2.33).
- (iii) If $x_n \rightharpoonup x$ and $u_n \rightarrow u$, then $\langle x_n | u_n \rangle \rightarrow \langle x | u \rangle$.

Corollary 2.52 (Key Characterization of Strong Convergence)

$$x_n \rightarrow x \iff [x_n \rightharpoonup x \text{ and } \|x_n\| \rightarrow \|x\|].$$

Intuition: weak convergence already gets every “linear measurement” right; what it can miss is the overall length (recall Example 2.32, where lengths stayed at 1 while the weak limit had length 0). Once you additionally know the lengths converge to the right value, there is no room left for the vectors to differ, and weak convergence upgrades to strong convergence.

Step-by-Step Proof of Corollary 2.52, with Numeric Check

Proof ($\Leftarrow$, the nontrivial direction)

Suppose $x_n \rightharpoonup x$ and $\|x_n\| \rightarrow \|x\|$. Expand, using bilinearity:

$$\|x_n - x\|^2 = \|x_n\|^2 - 2\langle x_n | x \rangle + \|x\|^2.$$

As $n \rightarrow \infty$: $\|x_n\|^2 \rightarrow \|x\|^2$ (given), and $\langle x_n | x \rangle \rightarrow \langle x | x \rangle = \|x\|^2$ (definition of $x_n \rightharpoonup x$, testing against $u = x$). Hence

$$\|x_n - x\|^2 \rightarrow \|x\|^2 - 2\|x\|^2 + \|x\|^2 = 0,$$

i.e., $x_n \rightarrow x$ strongly. ($\Rightarrow$ is immediate: strong convergence implies weak convergence and $\|x_n\| \rightarrow \|x\|$ by continuity of the norm.) ■

Contrast with Example 2.32, Numerically

For the orthonormal sequence (e_n) : $e_n \rightharpoonup 0$ but $\|e_n\| \equiv 1 \not\rightarrow \|0\| = 0$ — the norm-convergence hypothesis of Corollary 2.52 *fails*, which is exactly why $e_n \not\rightarrow 0$ strongly. This shows the hypothesis $\|x_n\| \rightarrow \|x\|$ cannot be dropped.

Python: Corollary 2.52 — Weak + Norm Convergence $\Rightarrow$ Strong

```
1 import numpy as np
2
3 # A sequence in  $\mathbb{R}^3$  converging strongly to  $x_{\text{star}}$ : weak convergence in
4 # finite dimensions coincides with strong convergence (Lemma 2.51(ii)),
5 # so we illustrate the mechanism of Corollary 2.52 explicitly.
6 x_star = np.array([1.0, -1.0, 2.0])
7 ns = np.arange(1, 400)
8 xn = [x_star + np.array([1.0, 0.5, -0.5]) / n for n in ns]
9
10 weak_err = [abs(np.dot(v - x_star, np.array([0.3, -0.7, 0.2]))) for v in xn]
11 norm_err = [abs(np.linalg.norm(v) - np.linalg.norm(x_star)) for v in xn]
12 strong_err = [np.linalg.norm(v - x_star) for v in xn]
13
14 print("n=399: weak-test error    =", weak_err[-1])    # -> 0
15 print("n=399: ||x_n||-||x|| err =", norm_err[-1])    # -> 0
16 print("n=399: strong ||x_n-x|| =", strong_err[-1])    # -> 0, as Cor. 2.52 predicts
17
18 # Contrast: orthonormal  $e_n$  has weak error  $\rightarrow 0$  but norm error does NOT  $\rightarrow 0$ 
19 N = 300
20 u_test = 1.0 / np.arange(1, N + 1)
21 for n in [10, 100, 300]:
22     e_n = np.zeros(N); e_n[n - 1] = 1.0
23     print(f"n={n}: <e_n|u_test>={np.dot(e_n, u_test):.4f} "
24           f"| ||e_n||-||0||={abs(np.linalg.norm(e_n) - 0):.4f} (stays 1, no strong cv)")
```

Lemma 2.53: Ostrowski's Theorem in Finite Dimensions

Lemma 2.53

Suppose $\mathcal{H}$ is finite-dimensional and $(x_n)_{n \in \mathbb{N}}$ is a **bounded** sequence with $x_n - x_{n+1} \rightarrow 0$. Then the set of **cluster points** of (x_n) is **compact and connected**. *Intuition: if consecutive terms get arbitrarily close together, the sequence cannot “jump” between two far-apart cluster regions without leaving a trail of intermediate points — so the cluster set cannot have gaps (this is Ostrowski’s classical theorem, Theorem 1.49, specialized to $\mathcal{H}$).* This is used later to analyze the asymptotic behavior of fixed-point iterations, where consecutive iterates naturally satisfy $x_n - x_{n+1} \rightarrow 0$.

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Gâteaux and Fréchet Derivatives: Intuition

Intuition

Both notions generalize the single-variable derivative $f'(x) = \lim_{h \rightarrow 0} (f(x+h) - f(x))/h$ to Hilbert-space-valued maps. **Gâteaux differentiability** only asks that the directional limit exists *separately* along each fixed direction y (a “one-line-at-a-time” derivative). **Fréchet differentiability** is stronger: it asks for one single linear approximation that works *uniformly* in every direction at once, exactly like the ordinary multivariable derivative in $\mathbb{R}^n$. Fréchet $\Rightarrow$ Gâteaux, but not conversely.

Definition 2.54 (Gâteaux Derivative)

Let $C \subset \mathcal{H}$, $T : C \rightarrow \mathcal{K}$, and $x \in C$ such that every segment from x stays in C for small steps. T is **Gâteaux differentiable** at x if there is $DT(x) \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ with

$$(\forall y \in \mathcal{H}) \quad DT(x)y = \lim_{\alpha \downarrow 0} \frac{T(x + \alpha y) - T(x)}{\alpha}.$$

$DT(x)$: the Gâteaux derivative, a continuous linear operator that best approximates T 's change along each direction y .

Definition 2.55 and 2.56: Fréchet Derivatives I

Definition 2.55 (Fréchet Derivative of $f : \mathcal{H} \rightarrow \mathbb{R}$)

Let $C \subset \mathcal{H}$ and $f : C \rightarrow \mathbb{R}$, with $x \in \text{int}(C)$. f is **Fréchet differentiable** at x if there exists $\nabla f(x) \in \mathcal{H}$ such that

$$\lim_{0 \neq y \rightarrow 0} \frac{f(x+y) - f(x) - \langle \nabla f(x) | y \rangle}{\|y\|} = 0.$$

The vector $\nabla f(x)$ is the **Fréchet gradient** of f at x . *Unlike Gâteaux differentiability, this requires the linear approximation to work uniformly in all directions, not just separately along each ray.*

Definition 2.55 and 2.56: Fréchet Derivatives II

Definition 2.56 (Fréchet Derivative for Operators)

For $T : C \rightarrow \mathcal{K}$ with $x \in \text{int}(C)$, T is **Fréchet differentiable** at x if there exists $DT(x) \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ such that

$$\lim_{0 \neq y \rightarrow 0} \frac{\|T(x+y) - Tx - DT(x)y\|}{\|y\|} = 0.$$

Higher-order derivatives are defined inductively: the **second Fréchet derivative** $D^2T(x) \in \mathcal{B}(\mathcal{H}, \mathcal{B}(\mathcal{H}, \mathcal{K}))$ satisfies

$$\lim_{0 \neq y \rightarrow 0} \frac{\|DT(x+y) - DT(x) - D^2T(x)y\|}{\|y\|} = 0.$$

Example 2.57: Quadratic Form with Linear Perturbation

Statement

Let $L \in \mathcal{B}(\mathcal{H})$, $u \in \mathcal{H}$, and

$$f : \mathcal{H} \rightarrow \mathbb{R}; \quad y \mapsto \langle Ly \mid y \rangle - \langle y \mid u \rangle.$$

Then f is **twice Fréchet differentiable** on $\mathcal{H}$ with $\nabla f(x) = (L + L^*)x - u$ and $\nabla^2 f(x) = L + L^*$.

Proof Sketch

Fixing x and setting $\phi(t) = f(x + ty)$, we have $\phi'(0) = \langle (L + L^*)x \mid y \rangle$. Comparing with the definition of the gradient, we find $\nabla f(x) = (L + L^*)x - u$. Since the gradient is linear in x , its derivative is the constant operator $L + L^*$, verifying $\nabla^2 f(x) = L + L^*$. ■

Example 2.59: Symmetric Bilinear Forms

Statement

Let $F : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{R}$ be a symmetric bilinear form bounded by $|F(x, y)| \leq \beta \|x\| \|y\|$ for some $\beta \in \mathbb{R}_+$. Set

$$f : \mathcal{H} \rightarrow \mathbb{R}; \quad y \mapsto \frac{1}{2}F(y, y) - \langle \ell \mid y \rangle$$

for some $\ell \in \mathcal{B}(\mathcal{H}, \mathbb{R})$. Then f is Fréchet differentiable on $\mathcal{H}$ with $Df(x) = F(x, \cdot) - \ell$.

Example 2.60: Squared Distance to a Point

Statement

Let $\mathcal{K}$ be a real Hilbert space, $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$, $r \in \mathcal{K}$, $x \in \mathcal{H}$, and

$$f : \mathcal{H} \rightarrow \mathbb{R}; \quad y \mapsto \|Ly - r\|^2.$$

Then f is **twice Fréchet differentiable** on $\mathcal{H}$ with $\nabla f(x) = 2L^*(Lx - r)$ and $\nabla^2 f(x) = 2L^*L$.

Intuition

This is the objective function in least-squares problems: minimize the squared error $\|Lx - r\|^2$ over x . The gradient points in the direction of steepest increase in error; the Hessian $2L^*L$ is always positive semidefinite, reflecting the convexity of the objective.

Lemma 2.61: Fréchet Differentiability Implies Gâteaux

Statement

If $T : C \rightarrow \mathcal{K}$ is Fréchet differentiable at x , then:

- (i) T is Gâteaux differentiable at x and the two derivatives coincide.
- (ii) T is continuous at x .

Intuition

Fréchet differentiability is stronger: it forces the same linear approximation to work uniformly in all directions. This automatically makes both the Gâteaux and Fréchet derivatives agree, and forces continuity.

Python: Gradient and Hessian of a Least-Squares Function

Example 2.60 gives $f(x) = \|Lx - r\|^2$ with $\nabla f(x) = 2L^*(Lx - r)$ and $\nabla^2 f(x) = 2L^*L$. We verify this numerically:

```
1  import numpy as np
2  rng = np.random.default_rng(3)
3
4  L = rng.normal(size=(3, 2))    # L in R^{3 x 2}
5  r = np.array([1.0, 2.0, -1.0])
6  x = np.array([0.5, -0.3])
7
8  # Objective: f(x) = ||Lx - r||^2
9  residual = L @ x - r
10 f_x = np.dot(residual, residual)
11
12 # Gradient: grad f(x) = 2 L^T (Lx - r)
13 grad_f = 2 * L.T @ residual
14
15 # Hessian: Hess f(x) = 2 L^T L
16 hess_f = 2 * L.T @ L
17
18 # Numeric gradient check via finite differences
19 eps = 1e-6
20 grad_fd = np.zeros_like(x)
21 for i in range(len(x)):
22     x_plus = x.copy()
23     x_plus[i] += eps
24     f_plus = np.dot(L @ x_plus - r, L @ x_plus - r)
```

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Summary: Core Concepts of Hilbert Spaces I

What is a Hilbert Space? (Recap)

A **Hilbert space** $\mathcal{H}$ is a complete inner-product vector space over $\mathbb{R}$: it has a notion of angle and length (via the inner product $\langle \cdot | \cdot \rangle$), is distance-complete (every Cauchy sequence converges), and is equipped with all the geometry of Euclidean space (orthogonality, projections, perpendicular decomposition). All results in this chapter remain valid in infinite dimensions — e.g., $\ell^2(\mathbb{N})$, $L^2(\Omega)$, H^1 Sobolev spaces — making Hilbert spaces the natural home for linear algebra and calculus in infinite dimensions.

Summary: Core Concepts of Hilbert Spaces II

Inequalities and Identities (The Backbone)

- **Cauchy–Schwarz:** $|\langle x | y \rangle| \leq \|x\| \|y\|$.
- **Parallelogram law:** $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$.
- **Polarization identity:** $4\langle x | y \rangle = \|x + y\|^2 - \|x - y\|^2$.
- **Orthogonality:** $x \perp y \iff \langle x | y \rangle = 0$ (abstract perpendicularity).
- **Strict convexity:** Balls in Hilbert space have no flat spots — the midpoint of two distinct equal-norm points is strictly shorter.

Summary: Core Concepts of Hilbert Spaces III

Operators and Functionals

- A **continuous linear functional** $f : \mathcal{H} \rightarrow \mathbb{R}$ is uniquely represented as $f(x) = \langle x \mid u \rangle$ for some $u \in \mathcal{H}$ (Riesz–Fréchet).
- Every continuous linear operator $T : \mathcal{H} \rightarrow \mathcal{K}$ has a unique **adjoint** T^* satisfying $\langle Tx \mid y \rangle = \langle x \mid T^*y \rangle$.
- **Monotone operators** (satisfying $\langle Tx \mid x \rangle \geq 0$) are the seed concept for convex optimization and monotone operator theory developed in later chapters.

Summary: Core Concepts of Hilbert Spaces IV

Weak vs. Strong Topology

- **Strong (norm) convergence** $x_n \rightarrow x$: $\|x_n - x\| \rightarrow 0$.
- **Weak convergence** $x_n \rightharpoonup x$: $\langle x_n | u \rangle \rightarrow \langle x | u \rangle$ for every $u \in \mathcal{H}$ (every “linear measurement” converges).
- In infinite dimensions, weak convergence is coarser: sequences may converge weakly (e.g., orthonormal sequences) without converging strongly.
- **Key fact:** the closed unit ball is *weakly* compact (Banach–Alaoglu–Bourbaki), recovering compactness in exchange for a weaker topology — essential for existence of minimizers.
- Weak + norm convergence $\Rightarrow$ strong convergence (Corollary 2.52): testing all linear functionals plus the length determines everything.

Summary: Core Concepts of Hilbert Spaces V

Differentiability

- **Gâteaux derivative:** directional limit along each ray separately; often discontinuous, limited utility.
- **Fréchet derivative:** uniform linear approximation in all directions at once; implies Gâteaux + continuity.
- Common examples: quadratic forms $f(x) = \langle Lx \mid x \rangle$ (gradient $\nabla f(x) = (L + L^*)x$), squared norms $f(x) = \|Lx - r\|^2$ (gradient $\nabla f(x) = 2L^*(Lx - r)$).
- The gradient ∇f is itself a continuous linear operator when f is twice Fréchet differentiable.

Next Chapter: Convex Sets and Their Geometry

Chapter 3: Convex Sets

With Hilbert space geometry now in hand, we pivot to *convex sets*: subsets C of $\mathcal{H}$ such that every line segment between two points in C stays inside C (i.e., $\alpha x + (1 - \alpha)y \in C$ for all $x, y \in C$, $\alpha \in [0, 1]$).

Key results:

- Every nonempty closed convex set has a unique **best approximation** to any point outside it (Theorem 3.16).
- Convex sets can be characterized by supporting hyperplanes, separating hyperplanes, and normal cones.
- Convexity + the weak topology interplay is crucial for the calculus of convex analysis developed in Chapters 4–5.